Memo to:

Subject: Factors effecting SIGMA Performance other than Internal Computing Speed.

I. Introduction:

As I mentioned in our conversations on this subject the past few days, it is certainly true that the many new and improved features of the SIGMA system improve its performance over the 704 quite significantly. It is also true, however, that the size of these effects depends drastically on the nature of the problem being done and even on the way in which a given problem is organized.

It is mainly because of this lack of quantitative measure that I have in the past avoided quoting performance figures based on them. Internal computing speed can be isolated and studied in great detail quantitatively. I think you will agree that the SIGMA Timing Simulator code on the 704 has been reasonably successful in this goal.

I am sorry that if my preoccupation with the quantitative details of the Simulator I may have given you the impression that internal computing speed is the whole story of the SIGMA system. It is certainly a key feature of the machine, the foundation of the system's performance; but it would be greatly handicapped without the other new features.

II. Formula Showing Performance Increase.

~~Because of their sensitivity to the nature of the problem being done, the following factors are very hard to substantiate. I claim no more than an educated guess on most of them.~~

Suppose we write the time that a given problem takes on the 704 as a sum of the times taken by internal computing and Input/Output:

$$T_{704} = T_c + T_o$$

The time taken by SIGMA can be written as

$$T_{SIG} = T_c/f_c + T_o/f_o$$

where f_c is the factor by which the calculating time is reduced and f_o is the factor by which the I/O time is reduced

The performance of SIGMA relative to the 704 is then

$$P = T_{704}/T_{SIG}$$

The factors f_c and f_o are themselves each made up of a number of terms, one way of breaking them down is as follows:

$$f_c = (f_{CALC})(f_{MEM})(f_{WDL})(f_{OP})(f_{OV})(f_{MULT})$$

$$f_o = (f_{RATE})(f_{FLEX})(f_{OV})(f_{MULT})$$

where the f's are ratios of SIGMA/704 for

f_{CALC} = Internal calculating speed

f_{MEM} = Effect of larger memory size

f_{WDL} = Effect of larger word length

f_{OP} = Effect of larger, more flexible opcode set.

f_{MULT} = Effect of new multiprogramming & organizational features.

f_{RATE} = Effect of faster word rates in I/O

f_{OV} = Effect of overlapping computing and I/O

f_{FLEX} = Effect of increased flexibility.

The factors f_{OV} and f_{MULT} appear in both I/O and compute as will be shown in the example.

If we define R as the ratio of I/O Time to compute Time in the 704 operations, the formula for performance may be written as:

$$P = \frac{1 + R}{\frac{1}{f_C} + \frac{R}{f_O}}$$

III. Estimates of the Performance Factors.

As was mentioned above, all the factors are sensitive to the nature of the problem being done, & therefore claim no more than an "educated guess" on most of them.

(1) f_{CALC} : The internal compute speed.

This is the subject